Project: Worked examples
Title: Heat exchanger (LMTD and ε-NTU)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates a single-phase heat exchanger with constant properties: the duty and outlet temperatures for a given area, or the area for a given duty. It covers counterflow, parallel flow, shell and tube, crossflow and one fluid at constant temperature.

WHAT IT CALCULATES	RELATION
Capacity rates	$C_h = \dot{m}_h \cdot c_{p,h}$, $C_c = \dot{m}_c \cdot c_{p,c}$, $C_r = C_{min}/C_{max}$; with one fluid at constant temperature ($C_r = 0$) that fluid's capacity rate is infinite, C_{min} is the other stream's and the constant stream leaves at its inlet temperature
Overall coefficient with fouling	$U_f = 1/(1/U_{clean} + R_f)$
Transfer units	$NTU = U_f \cdot A/C_{min}$; effectiveness $\varepsilon(NTU, C_r)$ for the arrangement
Heat exchanger duty	$\Phi = \varepsilon \cdot C_{min} \cdot (T_{h,in} - T_{c,in})$; outlets from the energy balance
Area for a required duty	$\varepsilon = Q_{req}/(C_{min} \cdot \Delta T_{max})$, NTU from the inverse relation, $A = NTU \cdot C_{min}/U_f$
Cross-check with LMTD	$\Phi = U_f \cdot A \cdot F_{corr} \cdot \Delta T_{lm}$, F_{corr} from R and P (one shell pass) or back-calculated

Basis: LMTD and effectiveness-NTU methods[1](§11.3–11.5, Tables 11.3–11.4, 7th ed.); Heat exchangers[2](Ch. 11, 5th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
arrangement	Flow arrangement	counterflow		assumption: Plate exchanger treated as pure counterflow.	any
mode	Mode	sizing (given duty)		assumption: Sizing a new exchanger.	any
$const_{side}$	Which side is at constant temperature (used only by the $C_r = 0$ arrangement)	hot side at constant temperature (e.g. condensing vapour)		default	any
m_h	Hot-side mass flow	1.2	kg/s	spec: Network connection agreement, primary flow	0.01 … 50 kg/s

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
cp_h	Hot-side specific heat	4190	J/(kg·K)	assumption: Water at about 70 °C.	500 … 10000 J/(kg·K)
$T_{h,in}$	Hot-side inlet temperature	80	°C	spec: Network supply temperature	-20 … 180 °C
m_c	Cold-side mass flow	1.5	kg/s	drawing: M-SCH-201 rev B	0.01 … 50 kg/s
cp_c	Cold-side specific heat	4180	J/(kg·K)	assumption: Water at about 50 °C.	500 … 10000 J/(kg·K)
$T_{c,in}$	Cold-side inlet temperature	40	°C	drawing: M-SCH-201 rev B	-20 … 150 °C
U_{clean}	Overall heat transfer coefficient (clean)	3000	W/(m²·K)	assumption: Typical clean plate exchanger, water/water.	20 … 8000 W/(m²·K)
R_f	Total fouling resistance	0.0001	m²·K/W	assumption: Treated closed circuits on both sides.	0 … 0.01 m²·K/W
dT_{min}	Minimum approach temperature difference	3	K	assumption: Practical approach for a plate exchanger.	0 … 50 K
Q_{req}	Required duty	120	kW	spec: Building peak heat load	0.001 … 100000 kW

3. Assumptions

A1. Single phase on both sides (except the $C_r = 0$ arrangement), constant properties, no heat loss to surroundings. (Incropera 7th ed. §11.3–11.4.)
A2. The crossflow both-unmixed effectiveness is the published approximate correlation, flagged on the sheet. (Incropera 7th ed. Table 11.3.)
A3. Specific heats are user records at the mean temperature of each stream. (Entered as reference records, which carry their source, edition and licence.)
A4. arrangement (Flow arrangement) is taken as counterflow. (Plate exchanger treated as pure counterflow.)
A5. mode (Mode) is taken as sizing (given duty). (Sizing a new exchanger.)
A6. cp_h (Hot-side specific heat) is taken as 4190 J/(kg·K). (Water at about 70 °C.)
A7. cp_c (Cold-side specific heat) is taken as 4180 J/(kg·K). (Water at about 50 °C.)
A8. U_{clean} (Overall heat transfer coefficient (clean)) is taken as 3000 W/(m²·K). (Typical clean plate exchanger, water/water.)
A9. R_f (Total fouling resistance) is taken as 0.0001 m²·K/W. (Treated closed circuits on both sides.)
A10. dT_{min} (Minimum approach temperature difference) is taken as 3 K. (Practical approach for a plate exchanger.)

4. Working

[12] $C_h = m_h * cp_h$
$= 1.2\ kg/s \times 4190\ J/(kg·K)$
$= 5028\ W/K$
Hot-side capacity rate

[13] $C_c = m_c * cp_c$
$= 1.5\ kg/s \times 4180\ J/(kg \cdot K)$
$= 6270\ W/K$
Cold-side capacity rate
[14] $C_{min} = \min(C_h, C_c)$
$= \min(5028\ W/K, 6270\ W/K)$
$= 5028\ W/K$
Minimum capacity rate
[15] $C_r = \min(C_h, C_c) / \max(C_h, C_c)$
$= \min(5028\ W/K, 6270\ W/K) / \max(5028\ W/K, 6270\ W/K)$
$= 0.80191$
Capacity ratio
[16] $U_f = 1 / (1 / U_{clean} + R_f)$
$= 1 / (1 / 3000\ W/(m^2 \cdot K) + 0.0001\ m^2 \cdot K/W)$
$= 2307.7\ W/(m^2 \cdot K)$
Overall coefficient with fouling
[18] $eps = Q_{req} / (C_{min} \cdot \Delta T_{max})$
$= 120000 / (5028 \times 40)$
$= 0.59666$
Effectiveness required
[19] $NTU = NTU(\varepsilon, C_r)$ inverse relation [1](7th ed. Table 11.4)
$= \varepsilon = 0.59666,\ C_r = 0.80191$
$= 1.2973$
Transfer units required
[20] $A_{req} = NTU \cdot C_{min} / U_f$
$= 1.2973 \times 5028 / 2307.7$
$= 2.8267\ m^2$
Required heat transfer area
[21] $UA = NTU \cdot C_{min}$
$= 1.2973 \times 5028$
$= 6523\ W/K$
Required UA
[22] $\Phi = \varepsilon \cdot C_{min} \cdot (T_{h,in} - T_{c,in})$
$= 0.59666 \times 5028 \times 40$
$= 120\ kW$
Heat duty
[23] $T_{h,out} = T_{h,in} - Q/C_h$
$= 80\ °C - 120000 / 5028$
$= 56.134\ degC$
Hot-side outlet temperature
[24] $T_{c,out} = T_{c,in} + Q/C_c$
$= 40\ °C + 120000 / 6270$
$= 59.139\ degC$
Cold-side outlet temperature
[25] $dT_c = T_{c,out} - T_{c,in}$
$= 59.139\ °C - 40\ °C$
$= 19.139\ K$
Cold-side temperature rise
[26] $T_{c,m} = (T_{c,in} + T_{c,out})/2$
$= (40\ °C + 59.139\ °C)/2$
$= 49.569\ degC$
Mean cold-side temperature

[27] $dT_{app} = \min(T_{h,in} - T_{c,out},\ T_{h,out} - T_{c,in})$
= 16.134 K
= 16.134 K
Approach temperature difference
[28] $dT_{lm} = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$, counterflow ends
= ΔT_1 = 20.861 K, ΔT_2 = 16.134 K
= 18.396 K
Log-mean temperature difference
[29] F_{corr} = 1 (pure counterflow, parallel flow or C_r = 0) [1](7th ed. §11.3)
= R = 1.247, P = 0.47847
= 1
LMTD correction factor
[30] $Q_{lmtd} = U_f \cdot A \cdot F \cdot \Delta T_{lm}$
= 2307.7 × 2.8267 × 1 × 18.396
= 120 kW
Duty by the LMTD method (cross-check)
[31] $u_{app} = dT_{min} / dT_{app}$
= 3 K / 16.134 K
= 0.18595
Approach utilisation (minimum / actual)

5. Results

SYMBOL	VALUE	DESCRIPTION
Φ	120 kW	Heat duty
$T_{h,out}$	56.13 °C	Hot outlet
$T_{c,out}$	59.14 °C	Cold outlet
dT_c	19.14 K	Cold-side rise
$T_{c,m}$	49.57 °C	Cold-side mean
cp_c	4180 J/(kg·K)	Cold-side specific heat
A_{req}	2.827 m²	Required area
C_r	0.80191	Capacity ratio
NTU	1.2973	Transfer units
eps	0.59666	Effectiveness
dT_{app}	16.13 K	Approach
UA	6523.0 W/K	Overall conductance U·A
dT_{lm}	18.40 K	Log-mean temperature difference
Q_{lmtd}	120 kW	Duty by the LMTD method
F_{corr}	1	LMTD correction factor
u_{app}	0.19	Approach utilisation (minimum / actual) (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Approach against minimum	3 K	16.13 K	0.19	

7. Applicability

WARNING records: tabulated values not taken from a reference record are cp_h, cp_c (Fluid properties (specific heats)). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
$const_{side}$: not entered, so the default of 'hot side at constant temperature (e.g. condensing vapour)' is used.

8. Verification

```
Template bs.heat-exchanger@1.5.0 Content hash 35bb6b8737222112 Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.
2. Çengel YA, Ghajar AJ. Heat and Mass Transfer: Fundamentals and Applications. 5th ed. McGraw-Hill Education; 2015.